Marketplace Revenue Leakage & Profitability Analysis

SQL-based analysis of profitability, discounts, returns, logistics costs and payment fees

Vishnu Narayanan Girishbabu
Business & Data Analytics Portfolio Project

MySQL | Data Analysis | Business Analysis

Business Problem & Objectives

Business Problem

- High marketplace sales do not necessarily translate into strong retained profitability. Discounts, product returns, fulfilment costs and payment processing fees can gradually erode margins even when topline revenue remains healthy.

Core Question

- Where is the marketplace losing revenue and profit, what is driving those losses, and where should management focus its attention?

Analysis Objectives

- Evaluate overall sales, costs and profitability
- Identify loss-making products and category-level performance
- Assess the financial impact of discounts and returns
- Analyse logistics and payment-fee leakage
- Identify customer and data-quality risks
- Translate findings into actionable business recommendations

Data → SQL Analysis → Leakage Diagnosis → Business Insights → Recommendations

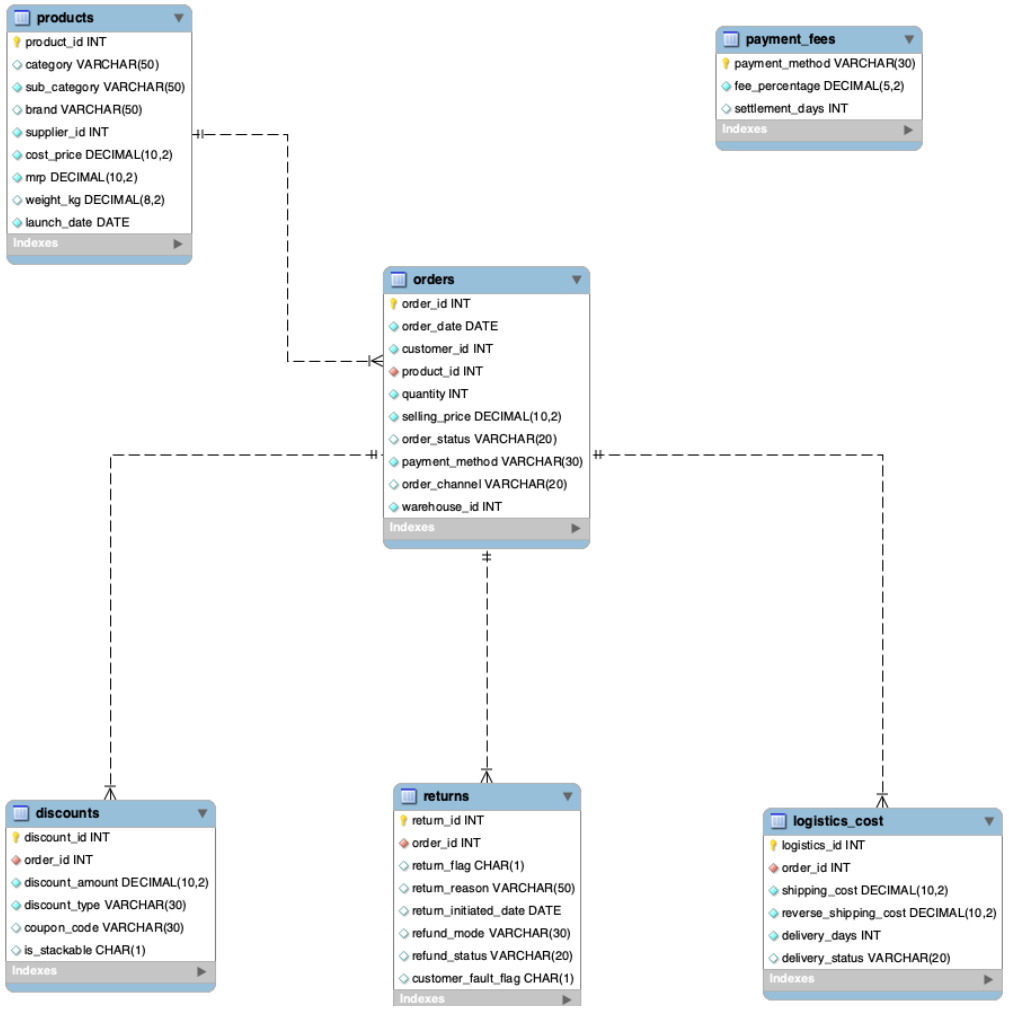
Data Model & Relational Structure

Data Model

Core Tables

- **Orders** — transaction-level order and sales data
- **Products** — product, category, brand and cost information
- **Discounts** — discounts applied to individual orders
- **Returns** — returned orders and return reasons
- **Logistics Cost** — shipping and reverse-shipping costs
- **Payment Fees** — processing fees by payment method

Orders act as the central transactional table connecting the marketplace's commercial and operational data.



Relational model used to connect sales, product, discount, return, logistics and payment data for profitability analysis.

Executive Profitability Snapshot

TOTAL SALES	PRODUCT COST	GROSS PROFIT
₹62.83M	₹30.32M	₹32.51M
		51.7% of sales

Business Insight
The marketplace demonstrates strong gross product profitability, but gross profit does not reflect the full economics of the business.

Profit is subsequently exposed to:
Discounts ▪ Returns ▪ Logistics Costs ▪ Payment Fees

Key Question

₹32.51M Gross Profit
↓
How much value is being eroded by commercial and operational leakage?

Category Performance

ELECTRONICS
₹12.00M Sales | ₹6.27M Gross Profit
Highest sales & gross profit

GROCERY
52.82% Gross Margin
Highest identified margin

FASHION
50.28% Gross Margin
Lowest identified-category margin

UNCATEGORISED
₹8.73M Sales | 13.9% of revenue
Classification gap

Business Insights

Electronics leads in sales (₹12.00M) and gross profit (₹6.27M), contributing 19.1% of total revenue.

Grocery achieves the highest gross profit margin (52.82%), despite generating lower sales than Electronics, Fashion and Home.

Fashion records the lowest margin among the five identified categories (50.28%).

Uncategorised products account for ₹8.73M in sales (13.9%), highlighting a significant product classification gap.

Business Implications

- Prioritise Electronics for further profitability and leakage analysis because of its high revenue contribution.
- Investigate Fashion's comparatively lower margin to identify potential pricing or product-cost inefficiencies.
- Resolve missing product categories to improve reporting accuracy and category-level decision-making.

Discount Usage & Effectiveness

Total Discount Value
₹5.73M

Discount-to-Sales Ratio
9.12%
Of gross order value

Orders with Discounts
7,417
33.71% of all orders

COUPON
₹1.54M
Largest identified promotional spend

UNSPECIFIED
₹1.52M
26.55% of total discount value

FESTIVAL
₹1.36M
23.67% of total discount value

BANK OFFER
₹1.31M
22.91% of total discount value

Key Business Findings

- Total recorded discounts amount to **₹5.73M**, equivalent to 9.12% of gross order value.
- Coupon discounts contribute the highest identified promotional expenditure at ₹1.54M.
- **26.55% of discount value (₹1.52M)** lacks discount-type classification, reducing promotional transparency.
- Multiple discount records exist for some orders, indicating potential discount stacking that requires validation.

Business Implications

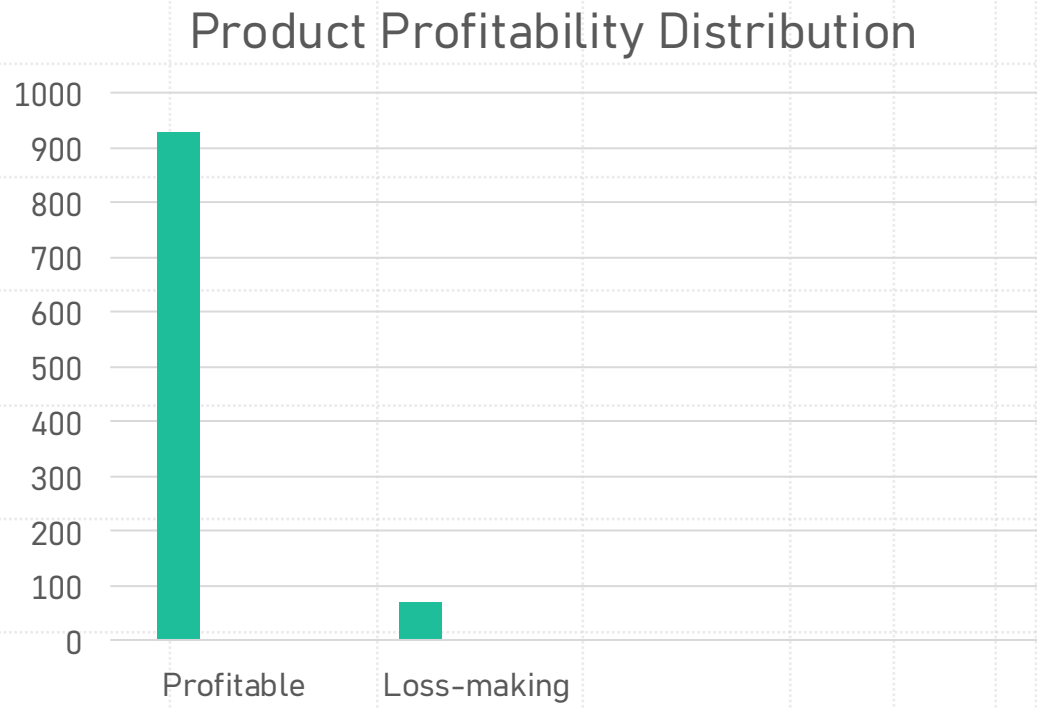
- Investigate unclassified discounts to improve promotional accountability.
- Review orders with multiple discounts for potential policy violations.
- Evaluate discount effectiveness against post-discount profitability before adjusting promotional strategies.

Product Profitability & Loss Risk

PRODUCTS ANALYSED	LOSS-MAKING PRODUCTS	LOSS-MAKING SHARE
1,000	71	7.1%

Key Business Findings

- **71 of 1,000 products (7.1%)** generated negative gross profit.
- Loss-making products can remain hidden when performance is viewed only at category or marketplace level.
- Product-level losses may reflect combinations of **selling price, procurement cost and discounting**.
- Product profitability varies materially even within otherwise profitable categories.



Business Implication

Introduce product-level profitability monitoring to identify persistent loss-makers before they materially erode category margins.

Returns & Financial Impact

RETURNED ORDERS	REVENUE LOST	GROSS PROFIT LOST
1,290	₹3.69M	₹1.93M

Rank	Return Reason	Revenue Lost
01	Unknown	₹714.7K
02	Damaged	₹636.4K
03	Quality Issue	₹628.7K
04	Customer Changed Mind	₹609.6K
05	Wrong Item	₹592.0K
06	Late Delivery	₹581.8K

- **Key Business Findings**
- **1,290 returned orders** were associated with approximately **₹3.69M in lost revenue** and **₹1.93M in lost gross profit**.
- Returns represent a material source of profitability erosion.
- **Unknown is the largest recorded return-reason group**, accounting for **242 returns and approximately ₹714.7K in lost revenue**.
- The high volume of unclassified returns limits management's ability to identify the underlying operational causes.

242 returns
Unknown reason

₹714.7K
Associated revenue loss

Key issue: The largest return-reason category is not actually classified.

Business Implication

Returns are both a profitability issue and a data-quality issue. Improving return-reason capture is necessary before management can effectively target root causes.

Logistics & Payment Cost Burden

Logistics Cost Exposure

₹1.19M
Total Logistics Cost

Key Finding
Orders where logistics cost exceeds **20% of order value** indicate potential fulfilment inefficiencies or uneconomical transactions.

Anomaly identified: One analysed order recorded ₹0 order value with ₹241 logistics cost.

Payment Processing Costs

₹1.10M
Total Payment Fees

Method	Fee Rate	Total Fees
COD	3.5%	₹437.3K
Card	2.1%	₹257.3K
Wallet	2.0%	₹251.7K
UPI	1.2%	₹152.1K

Highest fee burden → COD
Lowest fee rate → UPI

Business Implication: Fulfilment and payment costs create additional margin pressure beyond product economics. COD generates the highest payment-fee burden, while high logistics-to-order-value ratios highlight transactions requiring operational review.

Revenue Leakage Breakdown

Commercial and operational costs affect profitability through four major leakage channels.

DISCOUNTS
₹4.57M
Largest measured leakage component

RETURNS
₹2.95M
Second-largest measured leakage component

PAYMENT FEES
₹1.10M
COD carries the highest fee burden

LOGISTICS
₹0.95M
High-cost orders require investigation

Leakage Diagnosis

- Discounts and returns dominate the measured leakage profile, making commercial effectiveness and return management key areas for further investigation.
- Payment and logistics costs are individually smaller but remain recurring operational pressures on retained profitability.
- Leakage should not be treated as a single additive KPI without qualification, because the underlying SQL measures use different business definitions.

Executive Takeaway: Strong gross product profitability does not automatically translate into equally strong retained profitability. Management attention should shift from topline performance alone toward the mechanisms through which value is being eroded.

Customer & Data Quality Risks

High-Risk Customers
5.86%
Marketplace Average Return Rate

Customers with return rates above the marketplace average were classified as high-risk.

Several customers recorded **100% return rates**, indicating accounts that warrant further investigation.

Payment Data Quality
4,471 orders
INVALID Payment Method

INVALID is the largest recorded payment-method category by order count, indicating a significant payment-data classification issue.

Poor payment classification can distort channel-level analysis and reduce confidence in payment-cost decisions.

Return Data Quality
242 returns
Unknown Return Reason

Unknown is the largest return-reason group and is associated with approximately **₹714.7K in lost revenue**.

Missing return-reason information limits root-cause analysis and makes targeted corrective action more difficult.

Why This Matters

Reliable business decisions require reliable data. Customer-risk indicators may identify behavioural or operational issues, but incomplete payment and return classifications reduce the precision of profitability and root-cause analysis.

Management Recommendations

Priority	Recommendation	Action
01	Optimise Discount Strategy	Evaluate promotions using post-discount profitability, not sales uplift alone. Investigate the ₹1.52M of discounts with unspecified promotion types.
02	Reduce Return-Related Losses	Improve return-reason capture and investigate the major drivers of returned orders before implementing targeted interventions.
03	Control Fulfilment & Payment Costs	Review orders with unusually high logistics-to-order-value ratios and assess the commercial impact of high-cost payment methods, particularly COD.
04	Strengthen Profitability Monitoring	Track product-level profitability, persistent loss-makers, customer return behaviour and data-quality exceptions through recurring management reporting.

Management Focus: Protecting profitability requires coordinated action across pricing, promotions, returns, fulfilment, payment costs and data quality—not simply increasing sales.

Conclusion & Business Impact

The analysis shows that strong marketplace sales and gross product profitability can coexist with meaningful commercial and operational pressures on retained profitability.

PROFITABILITY

₹62.83M Sales | ₹32.51M Gross Profit

The marketplace is profitable at the product level, but aggregate performance masks **71 loss-making products**.

PROFIT EROSION

Discounts and returns represent the largest measured leakage components, while logistics and payment fees create additional recurring cost pressure.

CONTROL & DATA QUALITY

Unclassified discounts, INVALID payment methods and Unknown return reasons reduce analytical visibility and limit effective root-cause investigation.

Final Takeaway: Sustainable profitability requires management to look beyond revenue growth and systematically monitor product economics, promotional effectiveness, returns, fulfilment costs, payment costs and data quality.

SQL transformed transactional marketplace data into actionable profitability insights and management priorities.